

FREE RESOURCE

The 47 SQL Questions That Show Up in Every DE Interview

Based on 100+ real interviews tracked across
Microsoft, Walmart, Flipkart, PepsiCo & more.

12

Pages

47

Questions

3

Difficulty Levels

4+

Companies

WHAT'S INSIDE

- Easy / Medium / Hard — progressively harder
- Company-tagged so you know where each appeared
- Trick questions that decide offers at top companies
- Last page: where to get full answers + solutions

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How to Use This PDF

These 47 questions are drawn from 100+ DE interviews I personally tracked across Microsoft, Walmart, Flipkart, PepsiCo, and other product companies. They're organised into three tiers — Easy, Medium, Hard.

1 Start with Easy

Even if you feel ready. Interviewers use Easy questions to test precision — not just whether you know the answer, but whether your answer is exact.

2 Medium = most likely round

The bulk of system design + SQL rounds at Microsoft and Flipkart live here. Spend 60% of your prep here.

3 Hard = differentiator

These questions rarely appear word-for-word. What matters is that you can reason through them live. Practice talking out loud.

4 Company tags matter

If you're targeting Walmart, focus on Walmart-tagged questions first — same data domain, same interviewer culture.

5 Answers not included here

This PDF is your question bank. For full answers, reasoning, and 100 production scenarios — see the last page.

Easy Questions — Precision round. Get these perfect.

EASY Q01

Microsoft

What is the difference between WHERE and HAVING?

EASY Q02

Walmart

Explain the difference between INNER JOIN and LEFT JOIN.

EASY Q03

Flipkart

What does DISTINCT do and when would you use it?

EASY Q04

PepsiCo

How do you find duplicate rows in a table?

EASY Q05

Microsoft

What is the difference between DELETE, TRUNCATE, and DROP?

EASY Q06

Walmart

Write a query to find the second highest salary.

EASY Q07

Flipkart

What is a NULL value and how do you handle it in SQL?

EASY Q08

PepsiCo

How do you count rows per group? Give an example.

EASY Q09

Microsoft

What is the difference between UNION and UNION ALL?

EASY Q10

Walmart

Write a query to get all employees hired in the last 30 days.

EASY Q11

Flipkart

What is an index and why does it matter for performance?

EASY Q12

PepsiCo

Explain the ORDER BY clause with ASC/DESC examples.

Medium Questions (Part 1 of 2) — The bulk of most DE rounds.

MEDIUM Q13

Microsoft

Write a query using ROW_NUMBER() to rank employees by salary per department.

MEDIUM Q14

Walmart

What is the difference between RANK(), DENSE_RANK(), and ROW_NUMBER()?

MEDIUM Q15

Flipkart

How do you calculate a 7-day rolling average of daily sales?

MEDIUM Q16

PepsiCo

Explain CTEs vs subqueries — when would you prefer one over the other?

MEDIUM Q17

Microsoft

Write a self-join to find employees who report to the same manager.

MEDIUM Q18

Walmart

How do you pivot rows into columns in SQL? Give an example.

MEDIUM Q19

Flipkart

What is a correlated subquery? When is it a performance problem?

MEDIUM Q20

PepsiCo

Write a query to find customers who placed orders every month in 2024.

Medium Questions (Part 2 of 2) — Continued.

MEDIUM Q21

Microsoft

How does PARTITION BY differ from GROUP BY?

MEDIUM Q22

Walmart

Write a query to detect gaps in a sequential ID column.

MEDIUM Q23

Flipkart

How do you find the top 3 products per category by revenue?

MEDIUM Q24

PepsiCo

What is the difference between EXISTS and IN? Which is faster and why?

MEDIUM Q25

Microsoft

Write a query to calculate month-over-month revenue growth percentage.

MEDIUM Q26

Walmart

How do you handle slowly changing dimensions (SCD Type 2) in SQL?

MEDIUM Q27

Flipkart

Write a query to find users active in month N but not in month N+1.

MEDIUM Q28

PepsiCo

What is a materialized view and when should you use one?

Hard Questions (Part 1 of 3) — Differentiator territory.

HARD Q29

Microsoft

Design a SQL query to detect session boundaries from a clickstream table with timestamps.

HARD Q30

Walmart

Write a query to compute a cumulative retention curve across weekly cohorts.

HARD Q31

Flipkart

How would you write a query to find all pairs of products frequently bought together?

HARD Q32

PepsiCo

Explain query execution plans. What do you look for to diagnose a slow query?

HARD Q33

Microsoft

Write a recursive CTE to traverse an employee hierarchy tree.

HARD Q34

Walmart

How do you deduplicate a stream of CDC events with late arrivals in SQL?

HARD Q35

Flipkart

Design a SQL solution to calculate the median order value per region without PERCENTILE functions.

Hard Questions (Part 2 of 3) — Continued.

HARD Q36

PepsiCo

What are window frame clauses (ROWS vs RANGE)? Give an example where the difference matters.

HARD Q37

Microsoft

Write a query to identify 'whale' users: top 1% by spend, with at least 3 orders in 90 days.

HARD Q38

Walmart

How would you optimise a query that does a full table scan on a 500M-row fact table?

HARD Q39

Flipkart

Explain how Bloom filters and partition pruning work together in distributed SQL engines.

HARD Q40

PepsiCo

Write a query to find the longest streak of consecutive days a user was active.

HARD Q41

Microsoft

How do you handle fan-out joins in SQL that cause row multiplication? Give a fix.

HARD Q42

Walmart

Design a SQL-based idempotent pipeline: load data without creating duplicates on reruns.

Hard Questions (Part 3 of 3) — Continued.

HARD Q43

Flipkart

Write a query to assign each event to the closest prior campaign date per user.

HARD Q44

PepsiCo

What is the difference between a hash join, merge join, and nested loop join?

HARD Q45

Microsoft

How would you write a query to detect anomalies: days where sales deviate >2 std devs from the 30-day average?

5 Things Interviewers Never Tell You

1 They already know the answer.

They're watching HOW you think — whether you clarify scope, consider edge cases, and reason about trade-offs before you write a single line.

2 Silence kills.

If you're stuck, narrate your confusion. 'I'm thinking about whether to use EXISTS or IN here because of the subquery size...' is far better than silence.

3 Wrong answer + great reasoning = pass.

An incorrect query with a clear explanation of your intent often beats a correct query with no explanation. Interviewers at Microsoft confirmed this.

4 They expect you to ask questions.

Before writing SQL, ask: What's the table size? Is this a one-time query or a pipeline? Are there NULLs I should handle? It shows maturity.

5 The hard questions are about trade-offs.

For Hard-level questions, there is rarely one right answer. State your assumption, write one approach, then say what you'd do differently at scale.

Company Breakdown

Here's how the 47 questions map to companies I tracked:

Microsoft

12 questions

Strong on window functions, recursion, anomaly detection, and optimisation. Expect at least one Hard-level reasoning question.

Walmart

12 questions

Heavy on data pipeline thinking — idempotency, deduplication, and cohort analysis. Practical, warehouse-oriented SQL.

Flipkart

12 questions

Product analytics focus — user funnels, retention curves, A/B testing. Expects clean, readable SQL under time pressure.

PepsiCo

11 questions

More emphasis on joins, aggregations, and business logic. Solid fundamentals weighted over advanced tricks.

Trick Questions — What They're Really Testing

These aren't trick questions. They're precision filters.

Second highest salary

Tests: Do you know ROW_NUMBER vs LIMIT OFFSET? Will you handle ties correctly?
Most candidates write LIMIT 1 OFFSET 1 — which breaks with duplicate salaries.

NULL handling

Tests: Do you know that NULL != NULL in SQL? That COUNT(*) vs COUNT(col) behave differently? That WHERE col != 5 silently excludes NULLs?

UNION vs UNION ALL

Tests: Do you know UNION deduplicates (and is slower)? Will you choose UNION ALL when duplicates are impossible — because it shows you care about perf.

WHERE vs HAVING

Tests: Do you know HAVING filters AFTER aggregation and WHERE filters BEFORE?
Can you explain the order of SQL clause execution from memory?

Window function vs GROUP BY

Tests: Do you understand that window functions DON'T reduce rows? Can you explain PARTITION BY vs GROUP BY without confusing them?

You just went through 47 interview-grade SQL questions.
Now here's what most candidates never get access to:

Full answers. Production reasoning. 100 real scenarios.

"When Theory Ends" documents the exact scenarios, trade-offs, and reasoning I used to crack DE interviews at Microsoft — including full SQL walkthroughs.

Want answers + production scenarios behind these?

topmate.io/ankita_gulati/1920562

When Theory Ends · Rs.999 · Code: PRODUCTION100

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